

Name: _____

Date: _____

Directions. Complete the problems identified by your instructor. For multiple-choice problems, select one answer and give a concise justification; an unsupported letter is incomplete. Show enough work to make your reasoning clear.

A. Reconstruct and Challenge Natural-Language Arguments

A.1 Argument Structure

Identify how the claims fit together rather than merely listing them.

A.1.01

The team should archive a build log for every release. After all, archived build logs receive automatic timestamps, and any record with a trustworthy timestamp can be audited after a failure. Archived build logs can therefore be audited after a failure, while an auditable release process makes debugging more reliable.

- State the main conclusion, the intermediate conclusion, and every stated premise.
- Describe the support relationships and give one plausible bridge premise needed for the final recommendation.

A.1.06

Small commits reduce recovery time after a defect: focused version-control commits make it possible to reverse one change without discarding unrelated work, and selective reversal shortens recovery. The development team should therefore require each feature to be committed in focused steps.

- State the main conclusion, the intermediate conclusion, and every stated premise.
- Describe the support relationships and give one plausible bridge premise needed for the final recommendation.

A.2 Hidden Conditionals

Translate each rule exactly. Do not assume its converse.

A.2.03

Let M mean a member may vote, D mean dues are current, and A mean the member attended orientation. Rules: only members with current dues may vote; current dues are enough to vote unless orientation was missed; a member may vote if and only if dues are current and orientation was attended.

- Translate all three rules symbolically, treating 'unless orientation was missed' as requiring attendance here.
- State what the biconditional makes necessary and sufficient, then give the contrapositive of the first rule.

A.2.08

Let O mean an event is held outdoors, W mean the forecast is dry, and P mean a permit is approved. Rules: outdoor events require an approved permit; a dry forecast does not by itself guarantee an outdoor event; the event is held outdoors exactly when both the forecast is dry and the permit is approved.

- Translate the first and third rules and express the second as a non-implication.
- Classify W and P relative to O and give the contrapositive of the permit requirement.

A.3 Validity, Countermodels, and Repair

Rewrite each argument in numbered premise/conclusion form. If valid, name the rule; if invalid, give a true-premise/false-conclusion countermodel and a minimal repair.

A.3.05

If the cache is stale, then the integration test fails and the warning log records a mismatch. The integration test failed, and the warning log recorded a mismatch. Therefore, the cache is stale.

- Formalize the argument and evaluate it as directed.

A.3.10

The report is accurate exactly when both sensors agree and their calibration is current. The sensors agree, and an automatic check completed, but the premises do not say that this check establishes current calibration. Therefore, the report is accurate.

- Formalize the argument and evaluate it as directed.

B. Logical Reasoning I: Anatomy and Inference

B.1 Main Conclusion

Select the claim the author is ultimately trying to establish, then justify your choice by describing the role of the other claims.

B.1.02

The neighborhood clinic should begin offering Saturday appointments next month. Although its weekday schedule has short openings, those openings occur when many hourly workers cannot attend, and the last three evening clinics filled within a day. Weekend service was previously rejected because no nurse could cover it and no money had been allocated. A qualified nurse has now accepted the open position, and the existing budget covers the additional shift.

What is the main conclusion?

- The clinic should begin offering Saturday appointments next month.
- Short weekday openings do not adequately serve many hourly workers.
- Recent evening clinics provide evidence of unmet demand outside ordinary hours.
- The two practical obstacles that previously blocked weekend service have been removed.
- A qualified nurse can cover the proposed additional shift.

B.1.07

The department should waive Mira's late-registration fee. Its policy permits a waiver when a student took all reasonable steps to comply but was prevented by circumstances outside the student's control. Server logs show that the registration portal failed for the entire final hour, and Mira's saved form shows that she had completed every required step before the outage. Granting this waiver would apply the existing policy; it would not commit the department to excuse students who simply began late.

What is the main conclusion?

- The department should waive Mira's late-registration fee.
- Mira took all reasonable steps to comply before the deadline.
- The portal outage was outside Mira's control.
- The existing waiver policy applies to Mira's case.
- Granting Mira a waiver would not require waiving fees for students who merely began late.

B.2 Role of a Statement

Identify the argumentative function of the specified statement, not merely whether it is true.

B.2.04

Several engineers recommend closing the footbridge until it can be replaced, arguing that even targeted repairs will be expensive. The inspection report agrees about the cost but finds no structural instability and concludes that replacing two rail sections would restore the required safety margin. Closure would also force pedestrians onto a road with no sidewalk. The bridge should remain open during the targeted work, with access limited to one side.

The statement that repair will be expensive is presented as

- (a) the inspection report's reason for finding structural instability.
- (b) the author's reason for preferring complete replacement to targeted repair.
- (c) an intermediate conclusion derived from the lack of a sidewalk.
- (d) a qualification of the claim that the required safety margin can be restored.
- (e) a consideration favoring closure that the author grants but argues is outweighed.

B.2.09

A public evacuation alert is useful only if residents can understand it without consulting another source. The proposed alert uses three abbreviations familiar to emergency staff but undefined for the public, and its map labels use two of them. Many residents would therefore not understand the alert quickly enough. The agency should rewrite the alert in plain language and test it with readers who have no emergency-management training.

The claim that many residents would not understand the alert quickly enough is

- (a) an objection to testing the alert with untrained readers.
- (b) an intermediate conclusion derived from features of the draft and used with the usefulness condition to support revision.
- (c) a premise offered to explain what the abbreviations mean.
- (d) the general standard that defines when any alert is useful.
- (e) background that plays no part in the recommendation.

B.3 Point at Issue

Select the proposition one speaker accepts and the other rejects. Both positions must be textually supported.

B.3.01

Nara: The city should permit up to four food carts in the central plaza during lunch. Similar plazas became only slightly busier after adopting that limit, while nearby workers gained less expensive options. Ivo: Those plazas have wider paths. Even four carts would leave too little room here for wheelchairs during the noon rush, so the proposed permit should be denied.

Nara and Ivo disagree about whether

- (a) food carts can provide less expensive lunches than nearby restaurants.
- (b) the city should permit up to four food carts in the central plaza during lunch.
- (c) the central plaza has narrower paths than the comparison plazas.
- (d) wheelchair access should be considered when issuing permits.
- (e) a numerical limit can prevent a plaza from becoming crowded.

B.3.06

Amina: Every assigned video should receive accurate captions. Once the spoken content is available as text, each video will be fully accessible to students who cannot hear the audio. Joel: Captions are necessary and should be added, but a demonstration may communicate essential information only visually. Without descriptions of that information, captions alone will not make every video fully accessible.

Amina and Joel disagree about whether

- (a) accurate captions alone will make every assigned video fully accessible to students who cannot hear the audio.
- (b) every assigned video should receive accurate captions.
- (c) spoken content should be available as text.
- (d) some demonstrations communicate information visually.
- (e) visual information can be essential to understanding a demonstration.

B.4 Must Follow

Choose the statement that is logically guaranteed by the facts. Do not assume a converse, exclusivity, or an unstated relationship.

B.4.03

Some robotics-club members are first-year students assigned to the fabrication team. Every first-year fabrication-team member completed both safety training and a supervised tool demonstration. Anyone who completed the demonstration may reserve the beginner workbench, although completing it does not guarantee membership in the robotics club.

Which statement must follow?

- (a) At least one robotics-club member may reserve the beginner workbench.
- (b) Every robotics-club member completed a supervised tool demonstration.
- (c) Only first-year students may reserve the beginner workbench.
- (d) Some first-year students who are not club members completed the demonstration.
- (e) Every fabrication-team member completed safety training.

B.4.08

Every peer-reviewed article in the digital collection includes both a citation list and an author identifier. At least one article in the collection lacks a citation list, although every article has an author identifier. The collection also contains editorials, some of which have citation lists despite not having been peer reviewed.

Which statement must follow?

- (a) At least one article in the collection was not peer reviewed.
- (b) No article with an author identifier was peer reviewed.
- (c) Every article with a citation list was peer reviewed.
- (d) Exactly one article in the collection lacks a citation list.
- (e) At least one editorial lacks an author identifier.

B.5 Parallel Reasoning

Choose the argument with the same valid logical structure, ignoring subject matter.

B.5.05

Some archived memos concern the budget. Every budget memo is indexed, and every indexed memo is searchable. Therefore, some archived memos are searchable.

Which argument uses the same pattern of reasoning?

- (a) Some cedar trees are tall. Every cedar is a plant, and every plant needs water. Therefore, every tall thing needs water.
- (b) Some licensed pilots are instructors. Every instructor is trained, and some trained people fly at night. Therefore, some pilots fly at night.
- (c) Some fragile parcels are insured. Every insured parcel is tracked, and no tracked parcel is anonymous. Therefore, no fragile parcel is anonymous.
- (d) Some marked trails cross wetlands. Every wetland trail is mapped, and every mapped trail is inspected. Therefore, some marked trails are inspected.
- (e) Some safe bridges are inspected. Every inspected bridge is recorded, and every recorded bridge has an identifier. Therefore, every safe bridge has an identifier.

B.5.10

If the server is overloaded, requests slow down. Either the server is overloaded or the network is disconnected. Requests did not slow down. Therefore, the network is disconnected.

Which argument uses the same pattern of reasoning?

- (a) If the path is icy, travel slows. Travel is slow or the path is closed. The path is not icy, so the path is closed.
- (b) If the lock fails, the door opens. The lock fails or the alarm sounds. The door opens, so the alarm did not sound.
- (c) If the lamp is on, the room is bright. The lamp is off or the curtains are closed. The room is not bright, so the curtains are closed.
- (d) If the pump runs, water flows. Either the pump runs or the valve is closed. Water did not flow, so the valve is closed.
- (e) If the form is late, a fee applies. Either no fee applies or the form is late. A fee applies, so the form was late.

C. Logical Reasoning II: Assumptions and Evidence

C.1 Necessary Assumption

Select the assumption the argument requires. If it were false, the stated reasoning would fail.

C.1.02

During the month after a museum installed color-coded directional signs, the proportion of visitors who asked staff for directions fell sharply. Attendance remained about the same, and the museum did not reduce the number of staffed information desks. The facilities director concludes that the signs, rather than some unrelated change, produced most of the decline in requests.

Which assumption does the argument require?

- (a) Visitors who noticed the signs generally prefer reading them to speaking with museum staff.
- (b) No change unrelated to the signs accounts for most of the decline in requests for directions.
- (c) Each gallery used the same color for exits as every other gallery did.
- (d) A majority of all visitors noticed at least one of the new signs.
- (e) Reducing requests for directions allowed staff to complete other work.

C.1.07

At a city-sponsored cycling workshop, 186 volunteers from the Lakeview neighborhood completed a transportation survey. Nearly all favored converting one traffic lane into a protected bicycle lane. The workshop drew residents from every precinct, so the planning office concludes that Lakeview residents overwhelmingly favor the conversion.

Which assumption does the argument require?

- (a) The proposed bicycle lane costs less than every other transportation improvement under review.
- (b) Workshop volunteers are sufficiently representative of Lakeview residents in their views about the proposed conversion.
- (c) Each survey respondent lives in the precinct through which the proposed lane would pass.
- (d) Residents who oppose protected lanes understand the difference between protected and painted lanes.
- (e) The workshop attracted at least as many volunteers as the planning office expected.

C.2 Sufficient Assumption

Select the statement that, if added to the stated premises, makes the conclusion follow logically.

C.2.04

The town can fund either a footbridge or a fountain this year, but not both. The footbridge would provide the only safe pedestrian route across the stream during spring floods. The fountain would improve the central square and cost slightly less. The council should nevertheless fund the footbridge rather than the fountain.

Which assumption is sufficient to justify the conclusion?

- (a) A majority of residents who use the central square favor building the fountain.
- (b) Some residents who would cross the footbridge would also visit the central square.
- (c) The stream is difficult but not impossible to cross without a bridge during summer.
- (d) When exactly one of these projects can be funded, providing the safe spring crossing should take priority over the fountain's lower cost and benefit to the square.
- (e) The footbridge and fountain would be completed during the same construction season.

C.2.09

Route N and Route P connect the same trailhead to the visitor center. N is shorter, but it includes an unpaved segment; P is paved but has several street crossings. Neither route is currently closed. The guide nevertheless concludes that N will take less time to walk than P.

Which assumption is sufficient to justify the conclusion?

- (a) Both routes end at the same entrance to the visitor center.
- (b) Route P crosses more paved streets than Route N does.
- (c) Most visitors prefer a shorter route even when its surface is unpaved.
- (d) For N and P under their present conditions, whichever route is shorter also takes less time to walk.
- (e) Any route that is open to pedestrians can be walked in under an hour.

C.3 Diagnose the Flaw

Select the choice that most precisely describes the error in the argument's reasoning.

C.3.01

Attendance at the courtyard concert series rose by one third during the summer after organizers added more benches. Ticket prices and the number of performances did not change, and surveys say that patrons value having a place to sit. The organizers conclude that the added benches caused the attendance increase.

Which choice most accurately describes the flaw?

- (a) It infers that the benches caused the increase without excluding changes in performers, publicity, or other factors that could account for it.
- (b) It assumes that every additional attendee sat on one of the new benches rather than standing.
- (c) It overlooks the possibility that a few of the old and new benches remained empty at some performances.
- (d) It shifts from the total number of tickets sold to the average attendance at each performance.
- (e) It treats patrons' stated preference for seating as evidence that no patron values any other feature of the series.

C.3.06

Only editors who complete standards training and receive authorization from the managing editor may approve a public report. Dev is an editor and completed the training last week. Because the authorization list is not publicly posted, the department head concludes that Dev may now approve a public report.

Which choice most accurately describes the flaw?

- (a) It treats Dev's satisfaction of some stated necessary conditions as though it established the additional authorization needed for approval.
- (b) It assumes that Dev completed the training more recently than every editor already on the authorization list.
- (c) It overlooks the possibility that some internal reports need no approval before being circulated.
- (d) It infers that every person authorized to approve a report must have completed training from a claim only about Dev.
- (e) It treats the authorization list's not being public as evidence that no such list exists.

C.4 Strengthen

Select the statement that, if true, provides the strongest additional support for the argument's conclusion.

C.4.03

The four busiest unsheltered bus stops are exposed to direct afternoon sun. Riders can wait in nearby shops, but doing so sometimes causes them to miss a bus. The transit agency should install shaded shelters at those four stops because the shelters will make summer travel more usable without diverting money from essential service.

Which statement, if true, most strengthens the argument?

- (a) The four stops have different numbers of scheduled buses but similar proportions of delayed buses.
- (b) Many existing shelters display route maps that riders report finding useful.
- (c) Many riders avoid the stops on hot afternoons chiefly because shade is unavailable, and four shelters fit the existing improvement budget.
- (d) Routes serving two of the stops operate less frequently in winter than in summer.
- (e) Benches installed at two other stops last year were used during both sunny and cloudy weather.

C.4.08

A technical-support team's rate of incorrect case classifications fell after its members completed a checklist-training session. The software also received a minor visual update that week, but the classification rules displayed to employees did not change. The operations manager concludes that the training reduced the error rate.

Which statement, if true, most strengthens the argument?

- (a) The training presented examples in slides and repeated the checklist in a printed handout.
- (b) The team handles several categories of requests, and the decrease was largest in its most common category.
- (c) The same employees handled a comparable mix and volume before and after training, with no other change to classification procedures.
- (d) Employees who completed the session fastest did not necessarily have the lowest later error rates.
- (e) The manager chose the training date before learning when the visual software update would be released.

C.5 Weaken

Select the statement that, if true, most undermines the support offered for the argument's conclusion.

C.5.05

Every application reviewed by the compliance office receives a blue digital stamp that includes a date. Application T bears a current blue stamp and appears in the same portal used by the compliance office. The department therefore concludes that the compliance office reviewed T.

Which statement, if true, most weakens the argument?

- (a) Application T contains fewer pages than most applications reviewed this month.
- (b) The compliance office ordinarily reviews applications in the order in which the portal receives them.
- (c) Some stamped applications require revisions even after the compliance review is complete.
- (d) The date in T's blue stamp matches the date on which it was uploaded to the portal.
- (e) The portal automatically gives every uploaded application the same dated blue stamp before any compliance review occurs.

C.5.10

An application form contains several fields that first-time applicants often misunderstand. The agency proposes requiring every applicant to open an online tutorial before the submit button becomes active. Because the tutorial correctly explains every field, the agency concludes that this requirement will ensure that applicants complete the form correctly.

Which statement, if true, most weakens the argument?

- (a) The tutorial contains examples for the fields that applicants most often misunderstand.
- (b) Applicants may open the tutorial on a phone or computer before returning to the same form.
- (c) Staff can return incorrectly completed forms for revision after the application deadline.
- (d) The tutorial and form use identical labels for every corresponding field.
- (e) Applicants may close the tutorial immediately without reading it and then activate the submit button as usual.

D. Logical Reasoning III: Principles and Explanations

D.1 Principle Support

Select the principle that most strongly supports the stated judgment, and briefly justify the application.

D.1.02

A software cooperative promised users that it would promptly disclose defects that could seriously compromise their data. Engineers discover such a defect just before a major product demonstration. The cooperative's manager says it should disclose the defect now even though doing so may embarrass the organization.

Which principle most strongly supports the manager's judgment?

- (a) An organization should delay any announcement that might be misunderstood by the public.
- (b) An organization should honor a relevant public commitment even when honoring it creates a foreseeable reputational cost.
- (c) Technical information should be released only after every known defect has been repaired.
- (d) Organizations should disclose information only when disclosure improves sales.
- (e) A promise may be ignored whenever circumstances make compliance inconvenient.

D.1.07

A laboratory paper lists only researchers with faculty or student titles as authors. A technician designed a calibration procedure without which the reported measurements would not have been reliable. The lab director says the technician should also be credited as an author.

Which principle most strongly supports the director's judgment?

- (a) Every employee of a laboratory should be an author of every paper it produces.
- (b) Credit for a result should track a person's substantive intellectual contribution rather than merely the person's job title.
- (c) Only the person who writes the final draft should receive authorship credit.
- (d) Technical work should be credited only when it takes more time than the theoretical work.
- (e) Authorship should be assigned in order of seniority.

D.2 Resolve the Discrepancy

Select the fact that best allows both reported observations to be true, and explain how it resolves the apparent conflict.

D.2.04

Electronic gate counts show that fewer people entered the library this winter than last winter, yet the catalog recorded more new loans of physical books. Loan periods, renewal rules, collection size, and the definition of a new loan were unchanged.

Which fact best resolves the apparent discrepancy?

- (a) The library purchased reading-room chairs, but furniture deliveries were not counted by the electronic gates.
- (b) Many visitors who entered used wireless service without borrowing a physical book.
- (c) This winter was colder than the preceding autumn, though about as cold as last winter.
- (d) The library introduced outdoor lockers where patrons could initiate physical-book loans without entering the building.
- (e) The books borrowed this winter varied more in weight, but every book still counted as one loan.

D.2.09

A journal accepted a smaller percentage of submitted articles this year than last year, but it accepted a larger number of articles. The journal used the same definition of an accepted article, counted revised resubmissions in the same way, and completed decisions on all submissions in each year.

Which fact best resolves the apparent discrepancy?

- (a) Some accepted articles contained tables, but tables had no effect on how acceptances were counted.
- (b) The journal publishes online and increased the number of issues released during the later year.
- (c) At least one article was rejected in each year, consistent with both reported acceptance rates.
- (d) Submissions increased enough that the smaller accepted percentage represented a larger number of articles.
- (e) Editors read every accepted article before publication but also read many articles they later rejected.

D.3 Parallel Flaw

Select the argument whose reasoning contains the same flaw as the stimulus, and identify the shared invalid pattern.

D.3.01

If every archive box has been cataloged and the search index has finished refreshing, every record can be found through the public search page. Staff can now find every record through that page. Therefore, every box has been cataloged and the index has finished refreshing.

Which argument contains the same flaw?

- (a) If a classroom is occupied and its projector is active, the status panel glows. The panel glows, so the room is occupied and its projector is active.
- (b) If a badge is valid and the gate is online, the badge opens the gate. This badge is valid and the gate is online, so it opens the gate.
- (c) If the window is open or the smoke sensor activates, the alarm sounds. The alarm did not sound, so neither condition occurred.
- (d) The parcel is upstairs or downstairs but not both. It is not upstairs, so it is downstairs.
- (e) Every scanned and signed form is indexed. This form was scanned and signed, so it is indexed.

D.3.06

A voluntary online poll posted by opponents of a proposed stadium found that 88 percent of respondents opposed the stadium. Therefore, 88 percent of all city residents oppose it.

Which argument contains the same flaw?

- (a) A store invites buyers with strong opinions to review a new appliance; most respondents praise it, so the store concludes that most buyers are satisfied.
- (b) A registrar randomly samples students from every class year and uses the results to estimate campus commuting patterns.
- (c) Every attendee at a small workshop completed a survey, so the organizer accurately reports those attendees' views.
- (d) A survey's response rate was low, so its author reports the result with a wide margin of uncertainty.
- (e) A club polls all of its members about club meeting times and reports only the preferences of those members.

D.4 Method of Reasoning

Select the most accurate description of how the argument proceeds, and point to the feature that establishes that description.

D.4.03

A student argues that possession of a campus identification card guarantees entry to the archive. The archivist replies: 'An identification card is required, but visitors must also have a confirmed appointment. So the card alone does not guarantee entry.'

The archivist responds by

- (a) denying that an identification card is required for entry
- (b) offering an exception to the rule that all visitors need appointments
- (c) distinguishing a necessary condition from a sufficient one and identifying an additional requirement
- (d) arguing that appointments guarantee entry even without identification
- (e) questioning whether the student actually possesses a card

D.4.08

A researcher randomly selects 300 students, with the same proportion from each class year as in the university population. Sixty-four percent of the sample prefer later library hours. The researcher concludes that a majority of university students probably prefer later hours.

The argument proceeds by

- (a) deducing the conclusion from a rule that guarantees every student's preference
- (b) using one dissatisfied student as a counterexample
- (c) drawing a probabilistic generalization from a sample designed to represent the population
- (d) treating the conclusion as evidence for itself
- (e) showing that later hours are morally required

D.5 Complete the Argument

Select the claim that most logically completes the argument, and explain how the stated premises support it.

D.5.05

An archive's stated first priority is preventing the permanent loss of records that exist nowhere else; increasing access is important but secondary. Of this year's proposed projects, only digitizing the deteriorating field notebooks would preserve records with no duplicate. The other projects improve access to stable duplicated material. Therefore, ----.

Which choice most logically completes the argument?

- (a) every record lacking a digital copy should be digitized before any access project begins
- (b) the archive should stop accepting new records until the field notebooks are preserved
- (c) the deteriorating field notebooks must be the archive's most frequently consulted records
- (d) a preservation project is worthwhile only if it produces digital files for public access
- (e) digitizing the field notebooks should receive priority among this year's proposed projects

D.5.10

A committee agreed in advance to adopt a new scheduling system if a representative pilot reduced average delays without increasing the rate of scheduling errors. The completed pilot was representative: average delays fell by 18 percent, while the error rate remained unchanged. No additional adoption condition was specified. Therefore, ----.

Which choice most logically completes the argument?

- (a) the scheduling system will eliminate every delay after full adoption
- (b) the pilot proves that the system is better than the old one in every respect
- (c) the scheduling-error rate will decline after the system is adopted
- (d) the committee must add a cost condition before it can apply its prior decision rule
- (e) the committee should adopt the new scheduling system under its agreed rule

E. Integrated Evaluation and Revision

E.1 Causal Evaluation

Separate association from causation and say what additional evidence would matter.

E.1.02

Employees who used the optional standing desks completed more support tickets per week than employees who used ordinary desks. The standing desks therefore increased productivity.

- State the causal conclusion and the evidence offered for it.
- Explain why that evidence does not by itself establish causation and give one plausible alternative explanation.
- Describe one new finding that would strengthen the causal claim and one that would weaken it.

E.1.07

Users who turned off social-media notifications reported less stress one month later than users who left notifications on. Turning off notifications reduces stress.

- State the causal conclusion and the evidence offered for it.
- Explain why that evidence does not by itself establish causation and give one plausible alternative explanation.
- Describe one new finding that would strengthen the causal claim and one that would weaken it.

E.2 Apply a Principle

Choose the judgment that satisfies every condition in the principle. Then map each condition to a stated fact.

E.2.04

Principle: When officials have good evidence that two feasible policies would achieve the same important benefit, they should choose the one that restricts individual choice less, unless it would cost substantially more or create a distinct serious risk.

Which judgment most directly applies the principle?

- A city should choose a voluntary rebate because it restricts choice less, even though studies predict that it will achieve only half the pollution reduction of the feasible mandate.
- A city should choose a voluntary rebate whenever it is popular, without comparing effectiveness, feasibility, cost, or safety.
- A city should choose a voluntary rebate over a mandatory replacement program when controlled pilots show equal pollution reduction, both are feasible and similarly priced, and neither creates an additional safety risk.
- A city should choose a voluntary rebate that costs four times as much as the equally effective mandate because less restriction always controls.
- A city should choose a rebate and ignore evidence that its disposal method creates a serious fire risk absent from the equally effective mandate.

E.2.09

Principle: A rule adopted to prevent a specified harm should not prohibit conduct shown not to create or increase that harm merely because the conduct resembles harmful conduct in a way irrelevant to the rule's purpose.

Which judgment most directly applies the principle?

- A park rule aimed at preventing soil damage should allow every kind of equipment whose owner promises to be careful, even if tests show that some equipment increases erosion.
- A park should repeal its soil-protection rule because protecting soil is less important than allowing every visitor the same equipment choices.
- A park rule aimed at preventing soil damage should permit lightweight tripods that field tests show do not increase such damage, although they share three legs and a camera mount with the heavy rigs that do.
- A park may ban a lightweight tripod solely because it is expensive, even though expense is unrelated to soil damage, because rules need not track their announced purposes.
- A park rule should permit a heavy camera rig whenever it looks different from the equipment that originally caused erosion, even if it creates the same soil damage.

E.3 Integrated Op-Ed Analysis and Revision

Reconstruct and improve the argument while preserving what its evidence can genuinely support.

E.3.01

Some students share rooms or equipment and lack a dependable place to run long programs at home. A pilot last spring kept one lab open late, and students used it steadily. Any resource that is steadily used is worth its cost. Students who used the pilot also completed more assignments on time than students who did not. Thus the university should keep one computer lab open overnight during the final two weeks, since the pilot proves that overnight access improves academic performance.

- State the main conclusion and the strongest stated support.
- Identify one ambiguity and one inferential gap.
- Explain one alternative explanation for the reported association.
- Rewrite the final claim so it is appropriately qualified and supported.

E.3.06

The new introductory textbook has shorter chapters than the old one. Students in the first class to use it completed more assigned reading, and no instructor wants students to fall behind. The department should require the new textbook in every section because shorter books cause deeper learning.

- State the main conclusion and the strongest stated support.
- Identify one ambiguity and one inferential gap.
- Explain one alternative explanation for the reported association.
- Rewrite the final claim so it is appropriately qualified and supported.